Vibe-IC — All Steps: Phase → Stage → Step

Every step of the full flow, organised as **Phase → Stage → Step** with a **single continuous numbering 1 → 44** (starting from Stage 1's Spec-to-RTL). Phase 1's document-generation steps are labelled **D1–D5** (pre-flow, not counted in 1→44); two parallel tracks — **Analog A1–A9** and **Mixed-signal M1–M4** — run alongside.

Phase → Stage map

- **Phase 1** — Specification & documents: two entries (**Agent path** · **doc-gen path D1–D5**) + optional architecture-exploration front-ends
- **Phase 2** — RTL → synthesis: Stage 1 (RTL+verification) · Stage 2 (constraints+synthesis+DFT+handoff gate)
- **Phase 3** — Physical → Tapeout: Stage 3 (physical+sign-off) · Stage 4 (output+tapeout) · Stage 5 (manufacturing & test)
- **Parallel** — Analog A1–A9 · Mixed-signal M1–M4

Phase 1 — Specification & documents

Two entries; both produce the same L-series design documents that feed Phase 2:

- `phase1/input_prompt/` — free text / natural language → **Agent path**
- `phase1/input_doc/` — existing documents / structured YAML → **doc-gen path** (D1–D5)

Agent path (free-text input)

Agent	What it does
PM Agent	User-facing: turns natural-language requirements into design facts, asks one plain-

Agent	What it does
	language question per gap, hands off once confirmed.
IC Expert Agent	Behind the PM: reviews every layer with silicon expertise, fills sensible defaults, cross-layer consistency checks. Never user-facing.

Flow: user free text → PM Agent → IC Expert Agent → finalised L-series documents.

doc-gen path D1–D5 (existing-document input)

#	Step	What it does	Input	Output	Tools (EDA)	Programs
D1	Doc extraction → L1–L13	Ingest prompt/ docs into `input_doc/` and deterministically extract the core design layers.	user docs / prompt	`L1–L13` JSON	deterministic extractors	`phase1_al` skills: `data` `cmd-protoc` `adi-spec-g` `gen` · `test-` `waveform-g` `gen` · `integ` `test-cases` `gen` · `behav` `lab-calibr` `content-ge` `check` · `sch`
D2	Core design-layer docs L1–L13	Deterministic extraction of datasheet, FRS, register map, etc.	D1 plain text	`L1_DATASHEET … L13`	deterministic extractors	—
D3	Extended docs L14–L23	Protocol, timing, power intent (L21), skeletons.	L1–L13	`L14–L23` JSON	overlay extractor	—
D4	Protocol-class synthesis	Detect the IC's protocol class (81 classes) and synthesise class facts.	full input text	`ic_class` + protocol facts	is_ + _synth	—
D5						—

#	Step	What it does	Input	Output	Tools (EDA)	Programs
	Coverage report	Verify the input documents landed completely in the L docs.	input docs + L docs	parity / coverage report	parity reporter	

Architecture-exploration front-ends (optional, feed Step 1)

Front-end	What it does	Input	Output
architecture-explore	Design-space exploration (pipeline depth / parallelism / memory vs PPA, Pareto filter).	L docs + perf targets	architecture decisions (feed Step 1)
hls-c2rtl	C/C++/SystemC → RTL via HLS (open-source XLS, etc.).	C/SystemC model	RTL (verified from Step 2 onward)
SpinalHDL/ Chisel front-end	`eda_spinalhdl_gen` emits Verilog from Scala HDL.	SpinalHDL source	Verilog RTL

Front-end precedence (**artifact-driven**): existing RTL > C/SystemC model (hls-c2rtl) > SpinalHDL/Chisel > prompt-only (Step 1 spec-to-RTL). The available artifacts decide the path — never pick one speculatively.

Phase 2 — RTL → synthesis

Stage 1 — RTL generation & verification

#	Step	What it does	Input	Output	Tools (EDA)	Programs
1	Spec-to-RTL	Author synthesizable RTL from the L-series docs	L1–L23 docs	`rtl/*.v(.sv)` · coverage report	— (AI-authored from L docs;	skills: `spec`

#	Step	What it does	Input	Output	Tools (EDA)	Programs
		(SoC/CPU classes may take the IP-catalog reuse + glue path, e.g. Caravel-class harness platforms).			SoC via IP-catalog)	
2	🔁 Lint (RTL hygiene)	Static RTL style/bug checks; auto-fixable issues fixed first.	RTL	lint reports (hygiene / ROM-init)	Verilator lint	`rtl_hygien` `rtl_bug_re` `internal_v` skills: `rtl-`
3	🔁 CDC / RDC check	Clock-domain / reset-domain crossing safety.	RTL	CDC/RDC reports (crossing / async / reset-dep)	in-house CDC/RDC scan	`cdc_crossi` `cdc_async_` `reset_depe` skills: `cdc-`
4	🔁 Simulation (testbench + coverage)	Per-IC oracle testbench functional simulation (golden compares) + coverage measurement; when L21 declares power domains, the TB should cover power-state transition scenarios (no open-source UPF-aware sim — structural verification stays with M2).	RTL · L10 test cases	sim logs · results.xml · coverage report	iverilog/vvp · Verilator coverage `eda_simulate`	`testbench_` `l10_tb_con` `l12_tb_cov` skills: `testb`

#	Step	What it does	Input	Output	Tools (EDA)	Programs
5	🔁 Formal verification (assertions)	Prove key properties hold (`all_proved` requires the .sby + SymbiYosys evidence chain).	RTL · L3 constraints	`.sby` · formal results · full-stack TB results	SymbiYosys + smtbmc `eda_formal`	`assertion_` `bit_level_` `formal_pro` skills: `asse`
6	FPGA early prototype	Pre-synthesis behavioral prototype on FPGA.	RTL · board constraints	`.sof` · map report · FPGA verification audit	Quartus `eda_synth`	`fpga_test_` `debug_firs` `quartus_ma` `fpga_verif` skills: `fpga`

Stage 2 — Constraints, synthesis, DFT & handoff gate

#	Step	What it does	Input	Output	To
7	Constraint setup (SDC + PVT matrix)	Author timing constraints (SDC) + PVT corner matrix; power intent modelled via L21 and rendered to UPF (handoff artifact — open-source tools do not consume UPF; structural verification stays with M2).	L8 timing · L21 · PDK liberty	`*.sdc` · `pvt_matrix.json` · `<top>.upf` (optional, when L21 declares power domains)	— by `l2` scr ED
8	🔁 SDC validation (incl. derived-clock guard)	Validate constraint correctness, completeness, and exception (false_path/ multicycle) justification; manual ICG / register-divided clocks must declare a matching `create_generated_clock` (vacuous PASS without derived clocks — i.e. it passes automatically	SDC · L8 · RTL	SDC check report · `derived_clock_sdc.json`	—

#	Step	What it does	Input	Output	To
		when the condition does not apply: no divided clocks in the design means the check self-skips).			
9	Synthesis (Yosys → mapped netlist)	Synthesize RTL and technology-map (dfflibmap + abc -liberty) to the standard-cell netlist.	RTL · SDC · liberty	`synth/netlist.v` · area stats	Yo eda
10	🔁 Pre-layout STA (multi-corner)	Pre-layout static timing analysis (SS/TT/FF) + post-synth power preview (gate-level vectorless, default toggle rate, disclosed via `analysis_mode`; a different accuracy tier from Step 33's post-layout, optionally VCD-vector, analysis).	netlist · SDC · liberty	pre-PnR timing report + summary · `pre_pnr_power_preview.rpt`	Op eda
11	DFT insertion (scan chain + ATPG)	Insert scan chains + generate patterns (open-source Fault: scan + stuck-at ATPG + TAP; MBIST/LBIST/compression out of open-source scope).	netlist	scan netlist · ATPG coverage report	Fa (sc eda
12	Post-DFT optimization (resynth / buffering)	Re-optimise timing/area after DFT insertion.	scan netlist	`post_dft_netlist.v`	Yo
13	🔁 Equivalence check (RTL ≡ netlist)	Formally prove gate-level netlist ≡ RTL (LEC).	RTL · post-DFT netlist	LEC report	Yo
14	🔁 Synthesis handoff gate (pre-PnR Yosys audit)	Synthesis→PnR handoff QA: synth script + netlist audit (**open-source-Yosys specific**;	synth script · netlist	handoff audit reports	Yo ne

#	Step	What it does	Input	Output	To
		the synthesis stage's closing gate).			

Phase 3 — Physical design → Tapeout

Stage 3 — Physical design & sign-off

#	Step	What it does	Input	Outp
15	Floorplan + PDN	Chip floorplan + power delivery network; tapcell insertion (latch-up well-ties, SKY130 14 µm rule).	netlist · hardmacro LEFs (`pdk_local/` auto-included, via the IP integration check: LEF/GDS/Liberty alignment + corner coverage + L21 supply consistency; macro LEFs should carry obstruction layers)	`floor`
16	Clock planning	Clock-tree distribution strategy.	floorplan	`clock`
17	Placement (global + detailed)	Place standard cells.	floorplan · netlist	`place`
18	Spare-cell + ECO-prep insertion	Pre-place spare cells / ECO readiness so later fixes are metal-only (paired with Step 32 ECO: provisioned here, consumed there).	placed.def	`spare` cover
19	CTS (clock tree synthesis)	Build and balance the clock tree.	placed.def · clock plan	`post_` tree r

#	Step	What it does	Input	Outp
20	🔁 Post-CTS hold fixing	Repair hold violations after CTS (the runner repeats hold repair post-global-route).	post_cts.def	`post_`
21	Routing (global + detailed)	Complete all signal routing.	post_hold.def	`route` DRC r
22	Parasitic extraction (SPEF)	Extract post-route parasitics.	routed.def · tech LEF	SPEF
23	🔁 Post-route STA (multi-corner sign-off)	Sign-off timing with real parasitics (MMMC = per-corner loop, one report per corner).	netlist · SPEF · SDC · multi-corner liberty	post-r repor
24	🔁 IR drop	Power-grid voltage-drop analysis, PASS/FAIL vs the 5%-of-VDD budget (Caravel-class measured reference: worst <35 µV; the budget stays 5%-of-VDD).	routed.def (PSM)	IR rep budge
25	🔁 EM check (electromigration)	Current-density / metal-lifetime screen.	routed.def (PSM -enable_em)	EM re segme
26	🔁 Antenna check	Detect + repair process-antenna violations.	routed.def	anten
27	🔁 Signal integrity (crosstalk)	Crosstalk/noise screen (SPEF coupling-cap; advisory tier explicitly named).	SPEF	SI rep coupl
28	🔁 PERC / Reliability sign-off (ESD + latch-up + cross-domain)	Enforced sign-off: ESD pad-ring + discharge topology, latch-up well-tap, cross-voltage-domain protection; maps to the four PERC categories — netlist checks + netlist-driven layout checks (automated), current density + P2P resistance (named manual-review). Manual review is signed off by a senior physical-design / reliability engineer; criteria live in `perc_equivalent.json` (categories[].status=MANUAL_REVIEW + `review_criteria`: PDK Jmax tables, foundry ESD discharge-path P2P limit, Vhold>Vdd, L21 cross-domain	gate netlist · routed.def · L21 power intent · L3 ESD spec · step-24–27 reports	`perc_` · PERC verdi

#	Step	What it does	Input	Outp
		contract), results back-filled into checklist[].confirmed.		
29	Post-layout gate-level simulation (SDF)	Gate-level sim with SDF delays to confirm post-layout function (honest SKIP when no SDF re-sim ran).	gate netlist · SDF · TB	post-s
30	Post-layout SPICE verification	Transistor-level correlation for critical paths + analog blocks.	SPICE decks · SPEF	SPICE repor
31	🔁 Physical verification (DRC + LVS + ERC + density)	Sign-off physical rules; density here is **RULE COMPLIANCE** (KLayout per-layer CMP-window deck; execution verification → Step 34, optimization advisory → Step 35); LVS = Magic extraction + real netgen compare (macro-bearing designs — e.g. Caravel-class harnesses — may blackbox macros: Magic `lef write -hide` masks the macro to its interface shell, a netgen supplementary setup compares it as a same-name blackbox; the waiver basis = device-level match + KLayout cross-check, written by `signoff_waiver_emit` into the project's `waivers.json`, cross-referenced as reviewer to-dos by the Step 36 checklist's open_waivers).	GDS · gate netlist · PDK decks	sign-c ERC r
32	🔁 ECO (repair loop)	Engineering-change repair loop when sign-off finds issues (consumes the Step 18 spare cells first, for metal-only fixes).	sign-off reports	ECO l flag

Stage 4 — Output & Tapeout

#	Step	What it does	Input	Output
33	Power analysis (post-layout)	Full-chip power sign-off (post-layout vectorless OpenSTA report_power; optional VCD vector mode).	netlist · SDC · liberty (+ optional VCD)	power report (leakage+dynam analysis_mode)
34	Metal fill (density fill insertion)	Std-cell-row filler placement (white-space); density here	routed.def	`filled.def` · dens report

#	Step	What it does	Input	Output
		is **EXECUTION VERIFICATION** (the `metal_fill_density_check` gate owns the density verdict; rule compliance → Step 31, optimization advisory → Step 35).		
35	DFM screen (manufacturability)	Manufacturability screen: redundant-via ratio (single-cut fraction advisory) + density **OPTIMIZATION ADVISORY** (cross-references the Step 34 gate result only — never a duplicate FAIL); OPC/RET/SRAF/PSM as FOUNDRY_SIDE disclosure items (escalated to DESIGNER_COLLAB_REVIEW at ≤28nm; the node is derived by `dfm_screen_check` from the `input/pdk/liberty` filenames and recorded in the same `dfm_screen.json` — process_nm / advanced_node / foundry_side fields).	routed.def · density report	`dfm_screen.json` + foundry-side li
36	Tapeout checklist (final sign-off)	Item-by-item final confirmation (substance checks: DRC counts, evidence chains).	all sign-off reports	`tapeout_checkli`
37	GDSII output	Stream the foundry-deliverable GDSII (only when Step 31 PV is fully clean).	routed.def · merged GDS	sign-off `*.gds`
38	Foundry handoff (mask spec + WAT + scribe + corner vectors)	Foundry physical mask kit: mask spec + WAT plan + scribe PCM + corner ATE vectors (chip-specific; foundry-supplied fields named `PENDING_FOUNDRY_*` — tracked in the Step 36	GDS · netlist stats · L10 cases	`mask_spec.json` · `wat_plan.json` · S `corner_test_vec`

#	Step	What it does	Input	Output
		checklist and back-filled after the foundry replies).		
39	FPGA final sign-off (on-board test)	Final FPGA recompile + on-board attestation with hardware evidence.	RTL · board	final `.sof` · `on_board_pass.j`

Stage 5 — Manufacturing & test (post-fab; triggers on silicon receipt)

#	Step	What it does	Input	Output	Tools (EDA)	Pr
40	Fabrication (foundry, external)	Foundry mask-set + wafer fab (OPC/RET are foundry-side mask synthesis).	foundry handoff kit	mask/wafer intake attestations	external foundry	`ma`
41	Wafer sort / probe test	Wafer probing, good-die selection; yield independently re-derived vs target.	wafer lot · probe card	`wafer_sort_yield.json` · `wafer_map.csv`	ATE + probe card (external)	`wa`
42	Packaging (assembly)	wirebond / FC-CSP / WLCSP assembly.	good dies	`packaging_log.json`	assembly house (external)	`pa`
43	Final test (ATE + burn-in)	Post-package final test (functional + parametric + burn-in infant-mortality screen).	packaged units · ATE patterns	`final_test_yield.json` · `burn_in_results.json`	ATE (external)	`fi`
44	Reliability qualification (HTOL / FIT)	Long-duration HTOL qual (device-	HTOL chamber results	`htol_results.json` verdict	HTOL chamber (external)	`ht`

#	Step	What it does	Input	Output	Tools (EDA)	Pr
		hours / failures / FIT attestation; required for automotive/ medical grades; consumer MPW may stay dormant = DEFERRED, never blocks tapeout).				

> Out-of-flow lab steps (unnumbered): PFA/EFA (destructive FIB/ SEM/EMMI failure analysis) and silicon characterization (shmoo) — data originates from external equipment; the plugin owns the data-analytic root-cause layer (`wafer_map_pattern_classify`, yield-diagnostic).

Parallel tracks

Analog A1–A9 (parallel to Stages 1–3)

#	Step	What it does	Input	Output	Tools (ED
A1	Analog spec extraction	Extract per-block analog specs from the L docs.	L5 ADI spec	`spec.json`	determin extract
A2	Topology selection	Choose the circuit topology per spec.	spec.json	`topology.md`	AI topolog
A3	Netlist generation	Generate the SPICE netlist.	topology · PDK models	`<block>.sp`	xschem n `eda_xsche`
A4	Corner sweep (PVT)	PVT corner sweep + Monte-Carlo	`.sp` · PDK corner libs	`corner_results.json` (executed/derived counts)	ngspice (MC) `eda_spice`

#	Step	What it does	Input	Output	Tools (ED
		yield (`mc_yield_pct` ≥ 95% gate).			
A5	Analog layout	Complete the analog layout.	netlist	`layout.mag` / GDS	Magic lay `eda_analo`
A6	Block physical verification (per-block DRC + LVS)	Per-block DRC + LVS before merge (catches block-level errors top-level PV would mask).	block GDS · netlist	DRC clean / LVS match flags	Magic DR netgen LV
A7	🔁 Post-layout resimulation	Re-simulate with extracted parasitics; compare pre-vs-post specs (>10% degradation loops to A3).	layout parasitics · spec	`pre_vs_post.json`	Magic ext ngspice `eda_spice`
A8	Hardmacro generation (LEF + Liberty + GDS + Verilog)	Package the block for Stage-3 consumption (LEF should carry obstruction layers — Magic `lef write -hide` or abstract with obs — so top-level routing cannot enter the macro).	layout · characterisation	4-file hardmacro	Magic/abs character
A9	🔁 Co-simulation /	Mixed digital+analog simulation	hardmacro · digital RTL	cosim / HW measurement results	iverilog + co-sim

#	Step	What it does	Input	Output	Tools (ED
	HW verification	and hardware-in-the-loop verification.			eda_spice eda_simul

Mixed-signal M1–M4 (triggered when analog blocks exist)

Trigger timing: runs once **A8 hardmacros are complete** and **Stage 3 is near closure** (routed/GDS mergeable) — hardmacros must exist before floorplan, but M1's GDS merge waits for the digital side to finish routing. If a hardmacro slips until Stage 3 has entered Step 31, M1 waits for the final hardmacro GDS and Step 31's top-level LVS re-runs (incremental — scoped to the macro-interface change).

#	Step	What it does	Input	Output	Tools (EDA)	Program
M1	Top-level integration (A+D GDS merge)	Merge digital + analog GDS, place macros; run top-level LVS on the merged GDS (Magic extraction + netgen, macros blackboxed).	digital GDS · hardmacro GDS	`top_merged.gds` · merge/LVS report	KLayout merge + Magic/ netgen top-LVS	`mixed_sig` skills: `ana` `flow-orch`
M2	Power domain verification (level shifter / isolation)	Structural verification of cross-domain level-shifters / isolation (plus a DEF-level cross-domain	L21 · merged design	power_domain / level_shifter / isolation reports	structural check programs	`power_dom` `level_shi` `isolation` skills: `ir-`

#	Step	What it does	Input	Output	Tools (EDA)	Program
		sign-off check).				
M3	Mixed-signal verification (AMS co-sim)	AMS co-simulation + interface signal integrity.	merged design · cosim TB	cosim results · interface SI report	AMS co-sim	`mixed_sig` `mixed_sig` skills: `mix` `sim`
M4	Mixed-signal sign-off	Top-level verdict rolled up from M1–M3.	M1–M3 reports	`signoff.json`	rollup verdict	`mixed_sig` skills: `tap`

Totals

Phase	Stages	Steps
Phase 1 — Specification & documents	two entries (Agent · doc-gen) + architecture front-ends	D1–D5 + PM Agent · IC Expert Agent
Phase 2 — RTL → synthesis	Stage 1 · Stage 2	1–14
Phase 3 — Physical → Tapeout	Stage 3 · Stage 4 · Stage 5	15–44
Parallel	Analog · Mixed-signal	A1–A9 · M1–M4

44 sequential steps (Stage 1: 1–6 · Stage 2: 7–14 · Stage 3: 15–32 · Stage 4: 33–39 · Stage 5: 40–44), plus Phase 1 (Agent path & doc-gen path D1–D5) and the two parallel tracks (Analog A1–A9 · Mixed-signal M1–M4). Preflight: P0 (environment health check). Orchestrator `vibe_ic_one_shot_runner.py` runs Phase 1 → Phase 2 → Analog → Phase 3.

Out of scope (declined with reasons): designer-EXECUTED OPC/RET (mask synthesis is foundry-side — surfaced as FOUNDRY_SIDE items in Step 35 + noted at Step 40), commercial hardware emulators (FPGA path covers the intent), MBIST/LBIST/EDT compression (no open-source engines), BSR/BSDL, automatic clock gating (no characterized

ICG cell in sky130; manual RTL clock gating remains available — the SDC declaration of divided/gated clocks is guarded by Step 8's `derived_clock_sdc_required_check`), via-doubling/CAA (commercial DFM).

E1–E3 external-lab steps (reserved numbering, outside the 44; activated on automotive/medical grade upgrades):

#	Step	Notes
E1	PFA / EFA	FIB / SEM / EMMI failure analysis (external lab)
E2	Silicon characterization	shmoo plots · voltage/ frequency sweeps
E3	Temperature cycling / mechanical stress	JESD22-class environmental stress (external lab)

Appendix: measured wall-clock reference (for intuition)

Numbers come from REAL local sky130 open-source-flow runs (the `duration_s` fields of `reports/orchestrator/*_one_shot.json`; design sizes 0.5k–21k cells) — measured, never estimated. Durations vary with design, constraints and machine; nonlinear in cell count.

Stage	Measured range	Samples
Step 9 Synthesis (Yosys)	~0.4 s – 21 s	0.5k cells (lpc) → 20k cells (cv32e40p)
Steps 15–22 PnR (OpenROAD, full)	~3 s – 31 min	3.4k cells (subservient) → 21k cells (sha256)
Step 37 GDS write	~2 – 4 s	all samples
Step 31 DRC (KLayout sky130 deck)	~8 – 84 s	0.5k → 20k cells
Step 31 LVS (Magic + netgen)	this batch's samples took the light/skip path, so no representative figure; with real	—

Stage	Measured range	Samples
	macro compares (e.g. Caravel-class harnesses) it runs **minutes to hours**, scaling with macro count (a field run motivated the 4-hour timeout)	
Steps 40–43 manufacturing (fab/sort/pkg/final test)	external, weeks-scale	no local measurement

Licensing & IP: the whole flow relies on open-source tools only (Apache-2.0 project; commercial-tool firewall and output ownership: repo-root `README.md` §IP ownership, plus the DCO/patent pledge in `CONTRIBUTING.md`).

繁體中文版：`ALL_STEPS.zh-TW.md`.